

# Household Fuel Price Risk and the Uneven Protective Value of Domestic Net Zero: Evidence from Westminster, London

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## Abstract

Fuel poverty identifies current affordability hardship, but it does not by itself measure future household exposure to fossil-fuel price shocks. This paper develops the concept of household fuel price risk and applies it to Westminster, London, using property-level or property-archetype modelling. The empirical strategy estimates required energy demand, baseline energy burden, price-shock sensitivity, and the distributional effect of domestic low-carbon interventions. The central argument is that domestic net zero is not automatically a household risk-protection strategy. Retrofit, clean heat, solar PV, battery storage, smart tariffs, and flexibility reduce unequal exposure only when housing fabric, heating systems, tariff design, market reform, tenure, and local governance align.

**Keywords:** fuel poverty; energy burden; price risk; domestic retrofit; net zero; Westminster; energy justice

## 1 Introduction

This paper reframes household energy inequality from a static problem of current fuel poverty into a dynamic problem of uneven exposure to future fossil-fuel price volatility. Westminster is an analytically useful case because high average wealth can obscure fuel-price vulnerability among households facing high housing costs, inefficient dwellings, private rental constraints, leasehold governance, heritage restrictions, and limited access to low-carbon assets.

The central research question is: to what extent, and under what housing, market, and governance conditions, do domestic net zero interventions reduce unequal household exposure to fossil-fuel price volatility in Westminster?

The paper addresses three linked questions:

1. Which household-dwelling types face the largest required energy burden increase under fuel price shocks?
2. How do income, housing fabric, heating system, tenure, and vulnerability jointly produce fuel price risk?
3. How much risk reduction do retrofit, clean heat, PV, battery storage, smart tariffs, and integrated packages deliver, and for whom?

The contribution is to evaluate domestic net zero interventions as conditional household de-risking tools rather than as automatic bill-reduction devices.

## 2 Conceptual Framework

The paper separates three concepts that are often conflated. Fuel poverty identifies present affordability hardship under current prices and required energy needs. Energy burden measures

required or actual energy cost relative to residual household income. Household fuel price risk captures future exposure, sensitivity, and limited adaptive capacity under price-shock scenarios.

$$\text{Household fuel price risk} = \text{exposure} + \text{sensitivity} - \text{adaptive capacity}. \quad (1)$$

This distinction matters because decarbonisation is not the same as de-risking. Electrification can reduce emissions before it reduces household price risk where electricity prices remain linked to gas through wholesale market design. The protective value of heat pumps, PV, storage, smart tariffs, and flexibility therefore depends on the electricity-gas price ratio, tariff access, capital access, dwelling form, tenure, and governance.

### 3 Case and Data

The empirical case is Westminster, London. The preferred unit of analysis is the property or household-dwelling pair. Where household microdata are unavailable, the study uses EPC property records combined with LSOA/MSOA income and deprivation proxies to assign synthetic households or archetypes.

The current enriched property master contains 124,980 properties across 126 LSOAs and 26 MSOAs. It combines EPC attributes, spatial identifiers, MSOA after-housing-cost income, LSOA income deprivation, official LSOA fuel poverty rates, and LSOA income-distribution outputs from the latest verified pipeline.

The analysis uses required energy demand rather than only observed expenditure, because actual consumption may understate unmet need through underheating or self-rationing.

## 4 Methods

The modelling pipeline links building stock, household or archetype assignment, required energy demand, tariffs, fuel-price shocks, intervention scenarios, and spatial-distributional outputs.

### 4.1 Income and Required Energy Cost

The current income module estimates an LSOA-level lognormal income distribution:

$$Y_l \sim \text{Lognormal}(\mu_l, \sigma_l^2), \quad (2)$$

anchored to MSOA after-housing-cost income and shaped by LSOA income deprivation. Property-level required energy cost is estimated from SAP and floor area, then converted into burden probabilities such as the probability that required energy cost exceeds 10 percent of residual income.

### 4.2 Price-Shock Risk Metrics

The next modelling layer will define gas, electricity, combined fuel, standing-charge, and policy-mitigation scenarios. Candidate outputs include baseline energy burden, shock-after energy burden, burden uplift, threshold-crossing risk, affordability gap-at-risk, and Household Fuel Price-at-Risk.

### 4.3 Intervention Engine

Intervention scenarios will modify required energy demand, fuel structure, and tariff exposure. The core comparison is before versus after intervention, with results stratified by dwelling type, heating system, tenure, income distribution, EPC rating, and heritage or leasehold constraint.

## 5 Expected Outputs

The empirical outputs will include LSOA/MSOA maps, property-archetype summaries, distributional plots, and intervention comparisons. A central output is the de-risking dividend:

$$\text{De-risking dividend} = \text{Fuel Price-at-Risk}_{\text{before}} - \text{Fuel Price-at-Risk}_{\text{after}}. \quad (3)$$

The paper will also identify de-risking gaps: places and household-dwelling types where fuel price protection is most needed but low-carbon intervention is least accessible.

## 6 Discussion

The expected policy implication is that domestic net zero should be treated as a distributional household risk-protection agenda. Protection depends on the alignment of fabric retrofit, heating technology, distributed generation, flexibility, tariff design, electricity market reform, capital access, and tenure-sensitive governance.

In Westminster, this means paying particular attention to private renters, leasehold flats, HMOs, heritage-constrained dwellings, district-heating governance, social housing delivery pathways, and households whose vulnerability is masked by high neighbourhood average wealth.

## 7 Conclusion

Fuel poverty identifies current hardship; household fuel price risk identifies future exposure to price shocks. Net zero determines who receives protection only when domestic interventions are technically effective, financially accessible, and institutionally deliverable for the households most exposed.

The paper therefore asks not whether net zero reduces emissions, but whether, how, and for whom it credibly reduces household exposure to fossil-fuel price volatility.